

Car Parking Cost Calculation

Original Requirement (from Project Documentation)

Calculation of Car Parking

The following entries are required to calculate the total building assets construction costs:

Dropdown selection:

- Asset Group is the selection of the asset typology that this car parking entry will be related to
- Asset Typology is the car parking type i.e. Above_Ground, Basement, Podium or Free Standing
- Similar to building assets, base date, phase etc.

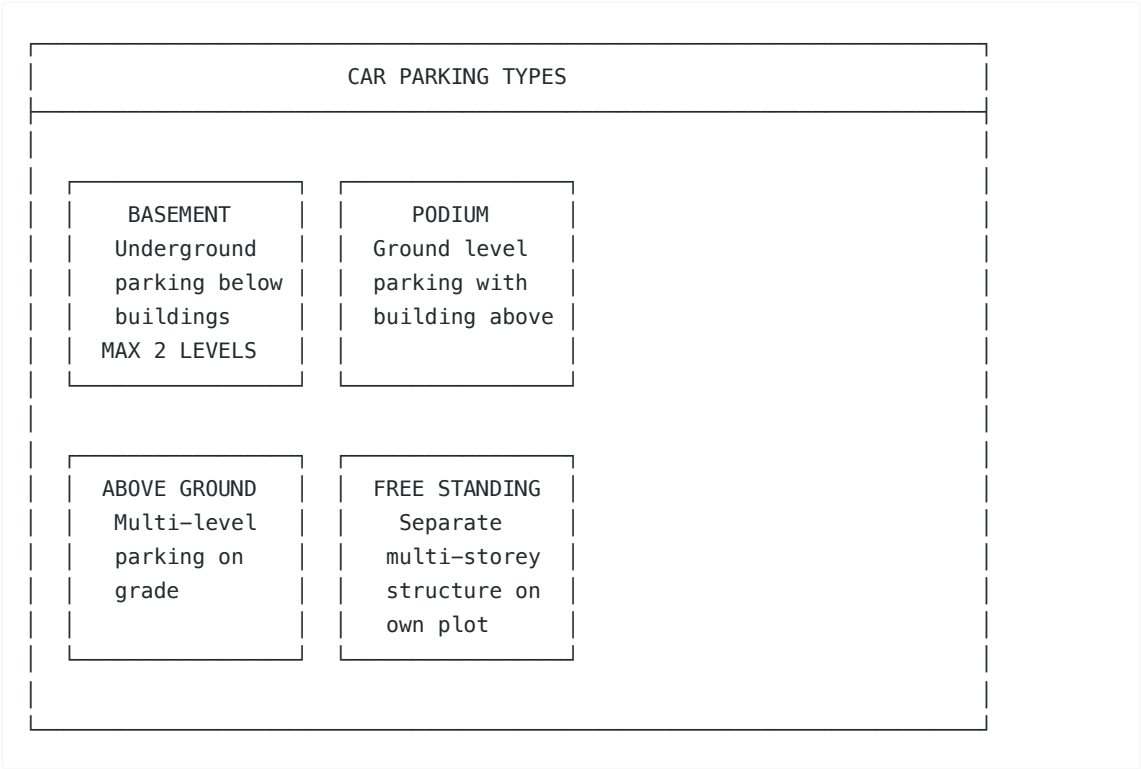
Free entry: Plot Area (in case of Free Standing parking it will make effect) - GFA/total parking area - Levels of each building, and levels (can't exceed 2 levels for basement due to the rate used).

Automated Calculations:

- In case of Free-Standing Parking, the total plot area is the plot area per building multiplied by the number of buildings. Similarly for total GFA. However, this is locked for basement and podium parking
- The FAR is locked/not needed
- The cost section will be calculated based on the selected car parking typology multiplied by parking area

Parking Types

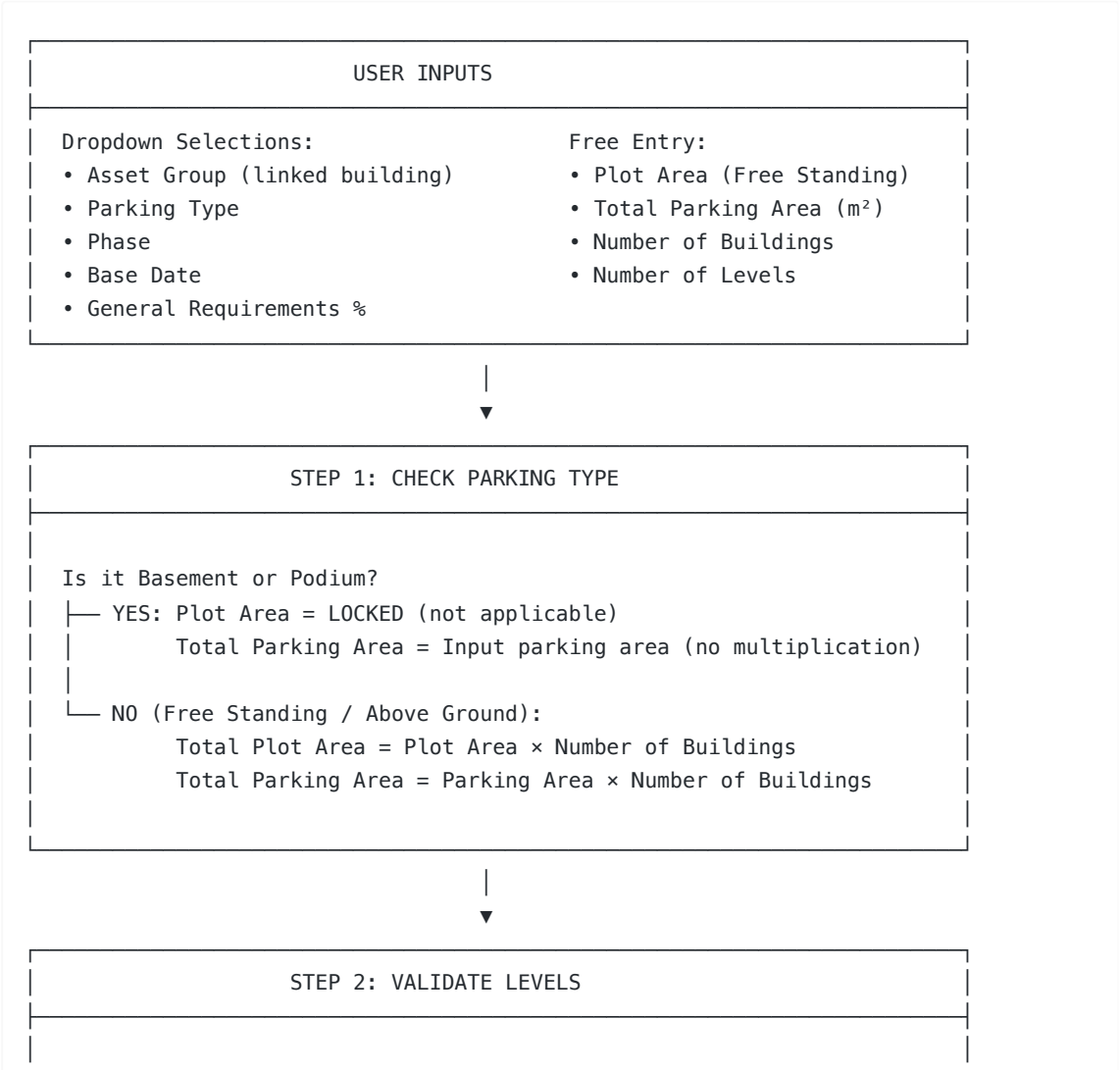
The platform supports four types of car parking:



Configuration by Type

Parking Type	Plot Area	FAR	Max Levels	Description
Basement	Not applicable	Not calculated	2	Underground parking below buildings
Podium	Not applicable	Not calculated	No limit	Parking integrated into building podium
Above Ground	Applicable	Calculated	No limit	Surface-level multi-level structure
Free Standing	Applicable	Calculated	No limit	Separate parking structure on own plot

Calculation Flow Diagram



Is it Basement parking?

└ YES: Check if Levels ≤ 2

 If Levels $> 2 \rightarrow$ ERROR: Basement cannot exceed 2 levels

└ NO: No level restriction



STEP 3: GET RATE FROM COST MODEL

SAR per m^2 = Retrieved from Cost Model based on Parking Type

Different rates for: Basement | Podium | Above Ground | Free Standing



STEP 4: COST CALCULATIONS

Net Build Cost = Total Parking Area $\times$ SAR per m^2

General Requirements Amount = Net Build Cost $\times$ (Gen Req % $\div 100$)

Total Cost = Net Build Cost $\times$ (1 + Gen Req % $\div 100$)



STEP 5: ADJUSTMENTS (Optional)

For Above Ground structures:

Facade Adjustment = Total Cost $\times$ (Facade Adjustment % $\div 100$)

Final Cost = Total Cost + Facade Adjustment

Apply Base Date Cost Factor if applicable



OUTPUT

- Total Plot Area (m^2) – if applicable
- Total Parking Area (m^2)

- | |
|--|
| <ul style="list-style-type: none">• Net Build Cost (SAR)• General Requirements Amount (SAR)• Total Cost (SAR)• Final Cost (SAR) |
|--|

Example Calculations

Example 1: Basement Parking

Input:

- Parking Type: Basement
- Total Parking Area: 5,000 m²
- Levels: 2
- General Requirements: 10%
- SAR per m² (from Cost Model): 2,500 SAR

Calculation:

- Total Parking Area = 5,000 m² (no multiplication for basement)
- Net Build Cost = 5,000 × 2,500 = **12,500,000 SAR**
- General Requirements = 12,500,000 × 0.10 = **1,250,000 SAR**
- Total Cost = 12,500,000 + 1,250,000 = **13,750,000 SAR**

Example 2: Free Standing Parking

Input:

- Parking Type: Free Standing
- Plot Area: 2,000 m²
- Parking Area per Building: 3,000 m²
- Number of Buildings: 2
- Levels: 4
- General Requirements: 10%
- SAR per m² (from Cost Model): 1,800 SAR

Calculation:

- Total Plot Area = 2,000 × 2 = **4,000 m²**
- Total Parking Area = 3,000 × 2 = **6,000 m²**
- Net Build Cost = 6,000 × 1,800 = **10,800,000 SAR**
- General Requirements = 10,800,000 × 0.10 = **1,080,000 SAR**
- Total Cost = 10,800,000 + 1,080,000 = **11,880,000 SAR**

Important Constraints

Maximum Basement Levels

Basement parking cannot exceed 2 levels.

This constraint exists because the cost rates in the model are calibrated for typical basement depths. Deeper basements would require different construction methods and costs that are not reflected in the standard rates.

Linking to Building Assets

Each car parking entry must be linked to a building asset group. This ensures:

- Proper cost allocation to developments
 - Accurate reporting of parking provision per asset
 - Correct aggregation in masterplan totals
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Parking Space Estimation

The platform can estimate parking spaces based on parking area:

Parking Type	Area per Space	Notes
Basement	30 m²	More circulation space needed underground
Podium	28 m²	Standard layout
Above Ground	28 m²	Standard layout
Free Standing	25 m²	More efficient layout possible

Estimated Spaces = Total Parking Area ÷ Area per Space